

3. Invariant measures

Haar measure

In many of the subsequent developments relations between subsets of $\mathbf{R}^n$ and m -dimensional planes in $\mathbf{R}^n$ will play a fundamental role. For example, we shall compare a set with its orthogonal projections on m -planes and with its intersections with m -planes. Often typical statements will hold not for all m -planes but for almost all m -planes. To make this “almost all” precise we need a measure on the space of all m -dimensional linear subspaces of $\mathbf{R}^n$. Also several of the proofs and statements of the results involve integration with respect to such a measure. In the same spirit we shall also use measures on the spaces of linear and affine isometries of $\mathbf{R}^n$. This is all part of the more general theory of invariant measures on homogeneous spaces, see e.g. Federer [3, 2.7]. However, from the general theory we shall only take for granted the well-known existence of a Haar measure on a compact topological group. Recall that G is a topological group if it is both a topological Hausdorff space and a group such that the group operations $(g, h) \mapsto gh$ and $g \mapsto g^{-1}$ are continuous.

3.1. Theorem. *If G is a compact topological group, there is a unique invariant Radon measure μ on G such that $\mu(G) = 1$. The invariance of μ means that for all $A \subset G$, $g \in G$,*

$$\mu(A) = \mu(\{gh : h \in A\}) = \mu(\{hg : h \in A\}).$$

For a proof see e.g. Halmos [1] or Munroe [1]. In fact, we shall prove the uniqueness in Theorem 3.4 for all the cases we shall need.

Note that any non-empty open subset U of G has a positive μ measure in the situation of Theorem 3.1. In fact, by the compactness we can cover G by finitely many sets $g_i U = \{g_i h : h \in U\}$ all having the same measure.

The uniqueness part of Theorem 3.1 is often a convenient way of proving formulas. For example, we obtain

$$(3.2) \quad \mu(A) = \mu(\{g^{-1} : g \in A\}) \quad \text{for } A \subset G,$$

since the right hand side also defines an invariant measure.

Uniformly distributed measures

We shall only need the existence of Haar measure in the case $G = O(n)$, the orthogonal group of $\mathbf{R}^n$; see below for the definition. It has the additional property of being a metric space with an invariant metric d , that is,

$$d(gh, gk) = d(hg, kg) = d(h, k) \quad \text{for all } g, h, k \in G.$$

It follows that $gB(h, r) = B(gh, r)$, and the invariance of μ implies that all balls with the same radius have the same measure. Such measures are called uniformly distributed.

3.3. Definition. A Borel regular measure μ on a metric space X is called *uniformly distributed* if

$$0 < \mu(B(x, r)) = \mu(B(y, r)) < \infty \quad \text{for } x, y \in X, \quad 0 < r < \infty.$$

We shall now give a simple proof due to Christensen [1] of the uniqueness of uniformly distributed measures.

3.4. Theorem. Let μ and ν be uniformly distributed Borel regular measures on a separable metric space X . Then there is a constant c such that $\mu = c\nu$.

Proof. Let g and h be the functions giving the μ and ν measures of the balls of radius r :

$$g(r) = \mu(B(x, r)), \quad h(r) = \nu(B(x, r)) \quad \text{for } x \in X, \quad 0 < r < \infty.$$

Let U be a non-empty bounded open subset of X . Clearly the limit $\lim_{r \downarrow 0} (\nu(U \cap B(x, r))/h(r))$ exists and equals 1 for $x \in U$. Hence by Fatou's lemma and Fubini's theorem

$$\begin{aligned} \mu(U) &= \int_U \lim_{r \downarrow 0} h(r)^{-1} \nu(U \cap B(x, r)) \, d\mu x \\ &\leq \liminf_{r \downarrow 0} h(r)^{-1} \int \nu(U \cap B(x, r)) \, d\mu x \\ &= \liminf_{r \downarrow 0} h(r)^{-1} \int_U \mu(B(y, r)) \, d\nu y \\ &= \left(\liminf_{r \downarrow 0} g(r)/h(r) \right) \nu(U). \end{aligned}$$

Interchanging μ and ν we obtain similarly

$$\nu(U) \leq \left(\liminf_{r \downarrow 0} \frac{h(r)}{g(r)} \right) \mu(U).$$

It follows that the limit $c = \lim_{r \downarrow 0} (g(r)/h(r))$ exists and $\mu(U) = c\nu(U)$ for every open set U . That $\mu = c\nu$ then follows by Theorem 1.10 (2) and the Borel regularity of μ and ν . $\square$

For example, the Lebesgue measure $\mathcal{L}^n$ and the Hausdorff measure $\mathcal{H}^n$ (see §4) are uniformly distributed Borel regular measures on $\mathbf{R}^n$. Hence $\mathcal{H}^n = c\mathcal{L}^n$.

Kirchheim and Preiss [1] proved, see also Kirchheim [1], that the support of a uniformly distributed measure on $\mathbf{R}^n$ is a real-analytic variety. A characterization of such measures seems to be a very difficult problem. For partial results, see Christensen [2] and Kirchheim and Preiss [1].

The orthogonal group

3.5. The orthogonal group $O(n)$ consists of all linear maps $g: \mathbf{R}^n \rightarrow \mathbf{R}^n$ preserving the inner product,

$$g(x) \cdot g(y) = x \cdot y \quad \text{for all } x, y \in \mathbf{R}^n,$$

or equivalently preserving the distance,

$$|g(x) - g(y)| = |x - y| \quad \text{for all } x, y \in \mathbf{R}^n.$$

(The equivalence is easy to check.) Then $O(n)$ is a compact subspace of the metric space of all linear maps $\mathbf{R}^n \rightarrow \mathbf{R}^n$ equipped with the usual metric

$$d(g, h) = \|g - h\| = \sup_{|x|=1} |g(x) - h(x)|.$$

With composition as a group operation it is also a topological group. We denote by θ_n its invariant measure with $\theta_n(O(n)) = 1$. Since d is invariant under composition, θ_n is uniformly distributed.

The members of $O(n)$ consist of rotations and rotations composed with a reflexion over some hyperplane. Another way to view them is to observe that they map orthonormal basis to orthonormal basis, and conversely given two orthonormal bases $u_1, \dots, u_n$ and $v_1, \dots, v_n$ of $\mathbf{R}^n$ one can define $g \in O(n)$ by setting $g(u_i) = v_i$ and extending linearly.

One of the basic properties of $O(n)$ is that it acts transitively on S^{n-1} : for any $x, y \in S^{n-1}$ there exists $g \in O(n)$ such that $g(x) = y$.

In the case $n = 2$, $O(2)$ is very simple. It consists of rotations around the origin and of rotations composed with the reflexion over the x -axis. Thus θ_2 can be identified with, for example, the normalized Lebesgue measure on $[-2\pi, 2\pi]$; the negative angles corresponding to the transformations containing reflexion. Alternatively it could be identified with the normalized length measure on the unit circle. In higher dimensions such an identification is no longer possible, but still the θ_n measures of many subsets of $O(n)$ can be reduced to the $(n-1)$ -dimensional surface measure on S^{n-1} , see Theorem 3.7.

We denote by σ^{n-1} the normalized ($\sigma^{n-1}(S^{n-1}) = 1$) surface measure on S^{n-1} . There are many ways to define it, for example as the normalized restriction of the $(n-1)$ -dimensional Hausdorff measure (see §4) $\mathcal{H}^{n-1}$ to S^{n-1} . Another way is to define it in terms of the Lebesgue measure via the formula

$$(3.6) \quad \sigma^{n-1}(A) = \alpha(n)^{-1} \mathcal{L}^n(\{tx : x \in A, 0 \leq t \leq 1\}), \quad A \subset S^{n-1},$$

where $\alpha(n) = \mathcal{L}^n(B(1))$. Then both σ^{n-1} defined by this formula and $\mathcal{H}^{n-1} \llcorner S^{n-1}$ are easily seen to be uniformly distributed measures on S^{n-1} , whence they are constant multiples of each other by Theorem 3.4.

3.7. Theorem. For any $x \in S^{n-1}$ and $A \subset S^{n-1}$,

$$\theta_n(\{g \in O(n) : g(x) \in A\}) = \sigma^{n-1}(A).$$

Proof. Defining $f_x : O(n) \rightarrow S^{n-1}$ by $f_x(g) = g(x)$, we have

$$\theta_n(\{g \in O(n) : g(x) \in A\}) = (f_{x\#}\theta_n)(A)$$

(recall 1.17). We have to show $f_{x\#}\theta_n = \sigma^{n-1}$. Since both give measure 1 for S^{n-1} , it suffices to show by Theorem 3.4 that $f_{x\#}\theta_n$ is uniformly distributed. Given $y, z \in S^{n-1}$ there is $h \in O(n)$ with $h(y) = z$. Then $h(B(y, r)) = B(z, r)$ for $0 < r < \infty$ and so by the invariance of θ_n

$$\begin{aligned} (f_{x\#}\theta_n)(B(z, r)) &= \theta_n(\{g : |g(x) - h(y)| \leq r\}) \\ &= \theta_n(\{g : |h^{-1} \circ g(x) - y| \leq r\}) \\ &= \theta_n(\{g : |g(x) - y| \leq r\}) \\ &= (f_{x\#}\theta_n)(B(y, r)). \end{aligned}$$

Therefore $f_{x\#}\theta_n$ is uniformly distributed. □

In Chapter 13 we shall need the following simple lemma.

3.8. Lemma. For any $x, y \in \mathbf{R}^n$, $x \neq 0$, and $\delta > 0$,

$$\theta_n(\{g : |x - g(y)| \leq \delta\}) \leq c\delta^{n-1}|x|^{1-n},$$

where c is a constant depending only on n . Moreover,

$$\theta_n(\{g : |x - g(y)| \leq \delta\}) = 0 \quad \text{if } ||x| - |y|| > \delta.$$

Proof. The last statement is trivial because $\{g : |x - g(y)| \leq \delta\} = \emptyset$ if $||x| - |y|| > \delta$. Suppose $||x| - |y|| \leq \delta$, $x \neq 0$ and $y \neq 0$. Then $|x - g(y)| \leq \delta$ implies

$$|x/|x| - g(y/|y|)| \leq 2\delta/|x|,$$

because

$$\begin{aligned} |x - g(|x|y/|y|)| &\leq |x - g(y)| + |(1 - |x|/|y|)g(y)| \\ &= |x - g(y)| + ||y| - |x|| \leq 2\delta. \end{aligned}$$

Thus we may assume $x, y \in S^{n-1}$. Using Theorem 3.7 we obtain

$$\begin{aligned} \theta_n(\{g : |x - g(y)| \leq \delta\}) &= \theta_n(\{g : g(y) \in B(x, \delta)\}) \\ &= \sigma^{n-1}(B(x, \delta) \cap S^{n-1}) \leq c\delta^{n-1}, \end{aligned}$$

where c is a constant depending only on n . □

The Grassmannian of m -planes

3.9. Let m be an integer with $0 < m < n$. We shall now introduce a natural measure on the so-called Grassmannian manifold $G(n, m)$ of all m -dimensional linear subspaces of $\mathbf{R}^n$. In the case $n = 2$, $m = 1$, this is trivial, since we can parametrize the lines through the origin in the plane by the angle they make with the positive x -axis, and then the one-dimensional Lebesgue measure on $[0, \pi]$ induces a measure on $G(2, 1)$. In fact, for the lines we can do almost the same in any $\mathbf{R}^n$, because every line through the origin pierces the unit sphere in exactly two points, and thus the surface measure on S^{n-1} induces a measure on $G(n, 1)$. From this we could also handle $G(n, n-1)$ by identifying any hyperplane with its orthogonal complement. However, for $2 \leq m \leq n-2$ there is no such simple and concrete method, and it is better to start from the Haar measure on $O(n)$; see however Exercise 6.

One way to get a natural metric on $G(n, m)$ is to identify $V \in G(n, m)$ with the orthogonal projection

$$P_V: \mathbf{R}^n \rightarrow V.$$

Then we can define for $V, W \in G(n, m)$

$$d(V, W) = \|P_V - P_W\|,$$

where $\| \cdot \|$ is again the usual operator norm for linear maps. With this metric $G(n, m)$ is compact. By simple linear algebra the action of $O(n)$ on $G(n, m)$ is distance-preserving:

$$d(gV, gW) = d(V, W) \quad \text{for } g \in O(n), \quad V, W \in G(n, m).$$

Also $O(n)$ acts transitively on $G(n, m)$: for $V, W \in G(n, m)$ there is $g \in O(n)$ such that $gV = W$. To see this, take orthonormal bases for V and W , complete them to orthonormal bases of $\mathbf{R}^n$, and choose $g \in O(n)$ which maps one of these onto the other.

Fixing $V \in G(n, m)$ we can now define a Radon probability measure $\gamma_{n,m}$ on $G(n, m)$ by

$$\gamma_{n,m}(A) = \theta_n(\{g : gV \in A\}) \quad \text{for } A \subset G(n, m).$$

In other words,

$$\gamma_{n,m} = f_{V\#} \theta_n \quad \text{with } f_V(g) = gV.$$

The invariance of θ_n implies also that $\gamma_{n,m}$ is invariant under $O(n)$, that is, for $g \in O(n)$, $A \subset G(n, m)$,

$$\gamma_{n,m}(gA) = \gamma_{n,m}(A) \quad \text{where } gA = \{gW : W \in A\}.$$

As before, the transitivity and distance-preserving property of the action of $O(n)$ on $G(n, m)$ imply that every $O(n)$ invariant Radon measure on $G(n, m)$ is uniformly distributed and consequently the invariant measure is unique up to a multiplication with a constant. In particular, this shows that $\gamma_{n,m}$ is independent of the choice of V .

From the uniqueness one also deduces

$$(3.10) \quad \gamma_{n,m}(A) = \gamma_{n,n-m}\{V^\perp : V \in A\} \quad \text{for } A \subset G(n, m),$$

where $V^\perp$ is the orthogonal complement of V . To prove this show that the right hand side is $O(n)$ invariant.

As mentioned above, the measures $\gamma_{n,1}$ and $\gamma_{n,n-1}$ can be reduced to the surface measure σ^{n-1} on S^{n-1} . More especially,

$$\begin{aligned}\gamma_{n,1}(A) &= \sigma^{n-1}\left(\bigcup_{L \in A} L \cap S^{n-1}\right), \quad A \subset G(n, 1), \\ \gamma_{n,n-1}(A) &= \sigma^{n-1}\left(\bigcup_{V \in A} V^\perp \cap S^{n-1}\right), \quad A \subset G(n, n-1).\end{aligned}$$

Again these formulas follow readily from the uniqueness.

Almost all further information we shall need about the measures $\gamma_{n,m}$ will be given in the following inequalities.

3.11. Lemma. *Letting $c = 2^n \alpha(n)^{-1}$ we have for any $x \in \mathbf{R}^n \setminus \{0\}$ and $0 < \delta < \infty$,*

$$\begin{aligned}\gamma_{n,m}(\{V : d(x, V) \leq \delta\}) &\leq c\delta^{n-m}|x|^{m-n}, \\ \gamma_{n,m}(\{V : |P_V x| \leq \delta\}) &\leq c\delta^m|x|^{-m}.\end{aligned}$$

Proof. Since $d(x, V^\perp) = |P_V x|$, the second inequality follows from the first and (3.10). To prove the first one, notice that

$$d(x, V) = |x| d(x/|x|, V),$$

which allows us to assume $x \in S^{n-1}$. Let

$$W = \{x \in \mathbf{R}^n : x_{m+1} = \cdots = x_n = 0\} \in G(n, m).$$

Then by (3.2), Theorem 3.7 and (3.6)

$$\begin{aligned}\gamma_{n,m}(\{V : d(x, V) \leq \delta\}) &= \theta_n(\{g : d(x, gW) \leq \delta\}) \\ &= \theta_n(\{g : d(g^{-1}x, W) \leq \delta\}) = \theta_n(\{g : d(gx, W) \leq \delta\}) \\ &= \sigma^{n-1}(\{y \in S^{n-1} : d(y, W) \leq \delta\}) \\ &= \sigma^{n-1}\left(\left\{y \in S^{n-1} : \left(\sum_{i=m+1}^n y_i^2\right)^{1/2} \leq \delta\right\}\right) \\ &\leq \alpha(n)^{-1} \mathcal{L}^n(\{z \in \mathbf{R}^n : |z_i| \leq 1 \text{ for } i \leq m, |z_i| \leq \delta \text{ for } i > m\}) \\ &= \alpha(n)^{-1} 2^m (2\delta)^{n-m} = \alpha(n)^{-1} 2^n \delta^{n-m}.\end{aligned}$$

□

3.12. Corollary. For $0 < s < m$ there is a constant c depending only on m, n and s such that for $x \in \mathbf{R}^n \setminus \{0\}$,

$$\int |P_V x|^{-s} d\gamma_{n,m} V \leq c|x|^{-s}.$$

Proof. Using Theorem 1.15 and Lemma 3.11 we compute

$$\begin{aligned} \int |P_V x|^{-s} d\gamma_{n,m} V &= \int_0^\infty \gamma_{n,m}(\{V : |P_V x|^{-s} \geq t\}) dt \\ &= \int_0^\infty \gamma_{n,m}(\{V : |P_V x| \leq t^{-1/s}\}) dt \\ &= \int_0^{|x|^{-s}} dt + \int_{|x|^{-s}}^\infty \gamma_{n,m}(\{V : |P_V x| \leq t^{-1/s}\}) dt \\ &\leq |x|^{-s} + c|x|^{-m} \int_{|x|^{-s}}^\infty t^{-m/s} dt \\ &= (1 + cs/(m-s))|x|^{-s}. \end{aligned} \quad \square$$

We shall further derive some, intuitively obvious information about generic relative positions of two linear subspaces of $\mathbf{R}^n$. We denote $G(n, 0) = \{0\}$ and $\gamma_{n,0} = \delta_0$ on $G(n, 0)$.

3.13. Lemma. Let k and m be integers such that $1 \leq k \leq n-1$, $0 \leq m \leq n-1$, $k+m \leq n$, and let $W \in G(n, k)$. Then

$$\gamma_{n,m}(\{V \in G(n, m) : V \cap W \neq \{0\}\}) = 0.$$

Proof. The lemma is clear for $n = 2$. We proceed by induction on n . Suppose the lemma holds for $n-1$ in place of n . We may assume $m \geq 1$. For any Borel set $A \subset G(n, m)$,

$$(1) \quad \gamma_{n,m}(A) = \int \gamma_{L^\perp, m-1}(\{U \subset L^\perp : L + U \in A\}) d\gamma_{n,1} L,$$

where $\gamma_{L^\perp, m-1}$ is the invariant measure on the Grassmannian of all linear $(m-1)$ -dimensional subspaces of $L^\perp$. This identity follows from the uniqueness of $\gamma_{n,m}$, since the right hand side defines an $O(n)$ invariant measure on $G(n, m)$. Evidently,

$$\gamma_{n,1}(\{L \in G(n, 1) : L \subset W\}) = 0,$$

and thus the integration in the above formula can be performed over the lines L with $L \not\subset W$. For any such L , the conditions $(L + U) \cap W \neq \{0\}$ and $U \subset L^\perp$ imply

$$L^\perp \cap (W + L) \cap U = (W + L) \cap U \neq \{0\}.$$

Hence by the induction hypothesis, as $\dim(L^\perp \cap (W + L)) \leq k$,

$$\begin{aligned} & \gamma_{L^\perp, m-1}(\{U : (L + U) \cap W \neq \{0\}\}) \\ & \leq \gamma_{L^\perp, m-1}(\{U : L^\perp \cap (W + L) \cap U \neq \{0\}\}) = 0. \end{aligned}$$

Actually, to use the induction hypothesis we also need $k \leq n - 2$, but otherwise $m - 1 = 0$ and the above statement is trivial. Integrating over the lines L with $L \not\subset W$ and taking in (1)

$$A = \{V \in G(n, m) : V \cap W \neq \{0\}\},$$

we obtain the desired result. $\square$

If $V, W \in G(n, m)$, then $V^\perp \cap W \neq \{0\}$ if and only if $P_V|W : W \rightarrow V$ is one-to-one. Thus we have

3.14. Corollary. *If $W \in G(n, m)$, then $P_V|W : W \rightarrow V$ is one-to-one for $\gamma_{n,m}$ almost all $V \in G(n, m)$.*

The isometry group

3.15. In addition to the linear isometries (rotations), we shall also consider the group $I(n)$ of general affine isometries (euclidean motions), which consists of maps $f : \mathbf{R}^n \rightarrow \mathbf{R}^n$ such that $|f(x) - f(y)| = |x - y|$ for $x, y \in \mathbf{R}^n$. They are exactly the maps composed of rotations and translations (see Hutchinson [1, 2.3]):

$$f = \tau_z \circ g \quad \text{where } g \in O(n), \quad z \in \mathbf{R}^n, \quad \tau_z(x) = x + z.$$

This representation is unique and we can metrize $I(n)$ by

$$d(f_1, f_2) = \|g_1 - g_2\| + |z_1 - z_2| \quad \text{where } f_i = \tau_{z_i} \circ g_i.$$

Then $I(n)$ becomes a locally compact, separable metric space, and we can define an invariant uniformly distributed Radon measure λ_n on $I(n)$ by requiring that for Borel sets $A \subset I(n)$

$$\lambda_n(A) = \int \theta_n(\{g : \tau_z \circ g \in A\}) d\mathcal{L}^n z.$$

Then for any non-negative Borel function φ on $I(n)$

$$\int \varphi d\lambda_n = \iint \varphi(\tau_z \circ g) d\theta_n g d\mathcal{L}^n z.$$

The affine subspaces

3.16. So far we have only considered the linear m -dimensional subspaces of $\mathbf{R}^n$, that is the m -planes through the origin. But we also want to use the space $A(n, m)$ of all m -planes of $\mathbf{R}^n$, the affine m -dimensional subspaces of $\mathbf{R}^n$. Every such m -plane T has a unique representation in the form

$$T = V_a = V + a \quad \text{where } V \in G(n, m), a \in V^\perp.$$

Here $V + a = \{x + a : x \in V\}$; thus V_a is the m -plane through a parallel to V . We metrize $A(n, m)$ by

$$d(V_a, W_b) = \|P_V - P_W\| + |a - b|,$$

and define a Radon measure $\lambda_{n,m}$ on $A(n, m)$ setting for Borel sets $A \subset A(n, m)$

$$\lambda_{n,m}(A) = \int \mathcal{H}^{n-m}(\{a \in V^\perp : V_a \in A\}) d\gamma_{n,m}V.$$

Here $\mathcal{H}^{n-m}$ is the $(n-m)$ -dimensional Hausdorff measure to be defined in Chapter 4, but at this point it is enough to know that its restriction to $V^\perp$ is simply a constant multiple of the $(n-m)$ -dimensional Lebesgue measure on $V^\perp$. For non-negative Borel functions φ on $A(n, m)$ we then have

$$\int \varphi d\lambda_{n,m} = \iint_{V^\perp} \varphi(V_a) d\mathcal{H}^{n-m}a d\gamma_{n,m}V.$$

The measure $\lambda_{n,m}$ is invariant under the transitive and distance-preserving action of $I(n)$ on $A(n, m)$. From this it follows that it is uniformly distributed.

We shall also denote $P_{V_a} = P_V + a$ for V_a as above.

Exercises.

1. Prove formula (3.2).
2. Prove that σ^{n-1} as defined in (3.6) is a uniformly distributed measure on S^{n-1} .
3. Show that the length measure on the spiral $\{(\cos t, \sin t, t) : t \in \mathbf{R}\}$ is uniformly distributed.
4. Show that $O(n)$ acts transitively on S^{n-1} and on $G(n, m)$.
5. Complete the details for the proof that $\gamma_{n,m}$ is uniformly distributed.
6. Denote by $L(v_1, \dots, v_m)$ the linear subspace spanned by the vectors $v_1, \dots, v_m \in \mathbf{R}^n$. Show that for $A \subset G(n, m)$,

$$\begin{aligned} \gamma_{n,m}(A) = \alpha(n)^{-m} \mathcal{L}^n \times \cdots \times \mathcal{L}^n (\{(v_1, \dots, v_m) \in (\mathbf{R}^n)^m : \\ |v_i| \leq 1, L(v_1, \dots, v_m) \in A\}). \end{aligned}$$